

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Forth Semester of B. Tech. (EC) Examination****May - June 2017****EC207.01 Control Systems****Date: 02.06.2017, Friday****Time: 10.00a.m. to 1.00p.m.****Maximum Marks: 70****Instructions:**

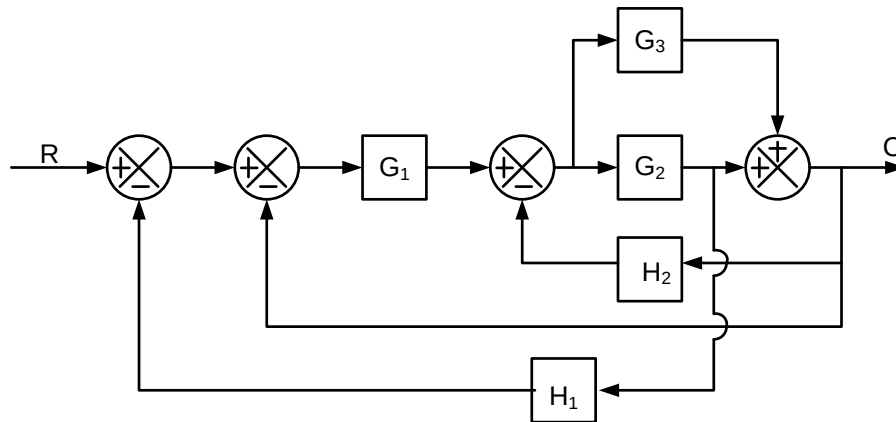
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

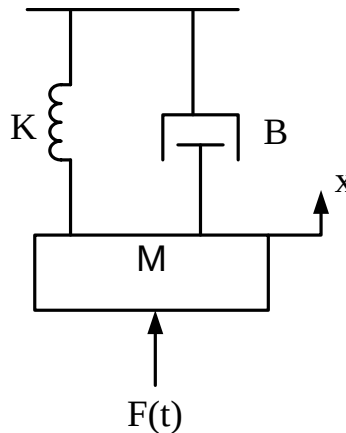
Q-1 Differentiate open and closed loop control systems with suitable examples. [07]

Q-2

- (a) Consider the block diagram shown in Figure 1. Using the block diagram reduction technique, determine transfer function $\frac{C}{R}$. [07]

**Fig.1 [Q-2a]**

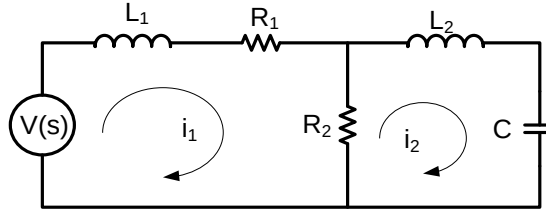
- (b) Draw the mechanical network for the system shown in Figure.2. Obtain an analogous electrical circuit based on Force-Current and Force-Voltage analogy. [07]

**Fig. 2 [Q-2b]**

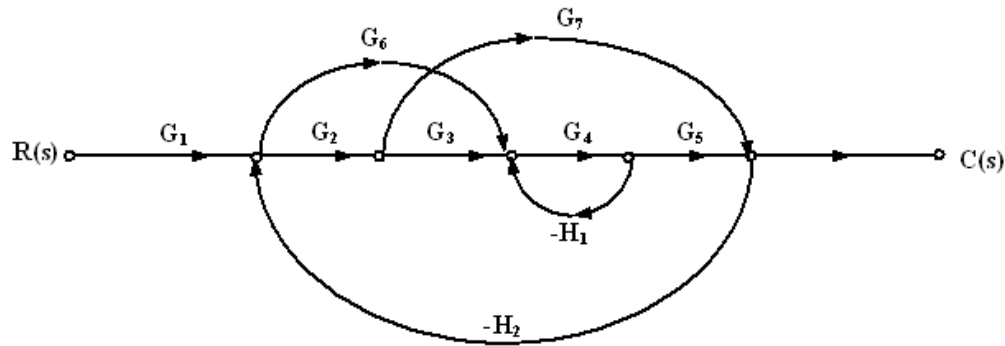
OR

Q-2

- (a) Obtain the transfer function $V(s)/I_1(s)$ of the network shown in Figure.3 [07]

**Fig. 3[Q-2a]**

- (b) Using Mason's formula, find the close loop transfer function for the block diagram shown in Figure 4. [07]

**Fig. 4 [Q-2b]**

Q-3

- (a) Derive the steady state error for following input signals: [07]
 (i) Unit step
 (ii) Unit ramp
 (iii) Unit Parabolic
- (b) Determine the stability of a system whose characteristics equation is given. Which roots make the system marginally stable. [07]

$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$

OR

Q-3

- (a) For the second order system with transfer function as given below obtain maximum percentage overshoot M_p and peak time t_p . [07]

$$G(s) = \frac{4}{s^2 + 2s + 4}$$

- (b) Sketch the bode plot for the transfer function given below. Determine gain cross over frequency, phase cross over frequency, Gain margin, Phase margin and stability of the system. [07]

$$G(s)H(s) = \frac{100}{s(s+1)(s+2)}$$

SECTION – II

Q -4 Discuss the Nyquist Stability Criterion. [07]

Q-5

- (a) Draw the root locus of the control system whose transfer function is following as K varies from 0 to ∞ . [07]

$$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$$

- (b) Sketch the bode plot for given transfer function. [07]

$$G(s)H(s) = \frac{512(s+3)}{s(s^2+16s+256)}$$

OR

Q-5

- (a) A feedback control system has an open loop transfer function [07]

$$G(s)H(s) = \frac{K}{s(s+3)^2}$$

Find the root locus as K is varied from 0 to ∞ .

- (b) Using Nyquist criterion, examine the closed loop stability of a system whose open loop transfer function is given by [07]

$$G(s)H(s) = \frac{10}{(s+1)(s+2)}$$

Q-6

- (a) A unity feedback system has [07]

$$G(s) = \frac{K(2s+1)}{s(4s+1)(s+1)^2}$$

Determine the value of K, if the steady state value of error to be less than 0.1 when input $r(t) = 1 + 5t$ is applied.

- (b) Define the following terms with respect to the bode plot. [07]

- (i) Gain Cross over frequency
- (ii) Phase Cross over frequency
- (iii) Gain Margin
- (iv) Phase Margin

OR

Q-6

- (a) List out rules to draw the root locus. [07]

- (b) For the unity feedback system with [07]

$$\frac{C(s)}{R(s)} = \frac{K}{(s+1)^3(s+4)}$$

Find the range of K for stability and also determine the frequency of oscillations when the system is marginally stable.
